

Design Document — Personal Homepage

Author: Ayush Nimeshkumar Patel **Course:** [CS5610 Web Development](#), Northeastern University
Project: Project 1 — Your personal home page **Live site:** <https://ayushp2207.github.io/ayush-homepage/> **Repository:** <https://github.com/ayushp2207/ayush-homepage> **PDF version:** [DESIGN.pdf](#)

1. Project description

What this is

A static personal homepage built with vanilla HTML5, CSS3 and ES6 modules. No backend, no framework, no component library, no build step. Bootstrap 5 is used for its grid only; every other style is hand-written.

The problem it solves

A résumé is a list of claims. "Kubernetes, PyTorch, LangChain" tells a reader that I have touched those things, but not *where*, *when*, or *how seriously*. Recruiters skim for keywords; engineers want evidence. The two audiences need the same facts presented at different depths, and a flat list of bullet points serves neither well.

So the organising idea of this site is **evidence, not adjectives**. Every skill claim is wired to the specific role, project or paper where I used it, and the site makes that wiring browsable rather than asking the reader to reconstruct it from prose.

Scope

Three pages:

Page	URL	Purpose
Home	<code>index.html</code>	Identity, the skill constellation, experience, education, toolkit
Projects	<code>projects.html</code>	Longer write-ups of projects and publications

Page	URL	Purpose
Built with AI	<code>built-with-ai.html</code>	The assignment's AI-generated page, used as a full GenAI disclosure

Non-goals

- No blog or CMS. Content lives in one JavaScript module and is edited by hand.
 - No contact form, because that would need a backend.
 - No dark/light toggle. The site commits to one dark palette and does it well.
-

2. User personas

Persona 1 — Priya, technical recruiter

- **Role:** University recruiter at a cloud infrastructure company.
- **Context:** Screening ~120 candidates for summer internships. Roughly 40 seconds per profile on the first pass, on a laptop with a dozen tabs open.
- **Goals:** Confirm graduation date, work authorisation timing, and whether the candidate has shipped anything at production scale.
- **Frustrations:** Portfolios that open with a full-screen animation and bury the graduation date three scrolls down. Having to open a PDF to learn anything concrete.
- **What she needs from this site:** Name, current role, graduation date and availability visible without scrolling. One click to email.

Persona 2 — Daniel, engineering hiring manager

- **Role:** Senior engineer who will run the technical interview.
- **Context:** Already read the résumé and is deciding whether to spend an hour on a phone screen. Reads carefully, and is sceptical of inflated claims.
- **Goals:** Find out whether the candidate's "multi-agent systems" experience is a tutorial project or real work; see actual code.
- **Frustrations:** Skill bars showing "Python 90%". Project cards with a title, a logo and no explanation of what was actually built or how hard it was.
- **What he needs from this site:** The ability to pick a technology and immediately see every place it was used, with enough detail to form a question. A link to source.

Persona 3 — Meera, fellow student and peer reviewer

- **Role:** Classmate assigned to review this submission.
 - **Context:** On a phone, between classes, working through a rubric.
 - **Goals:** Check the rubric items — semantic markup, responsive layout, alt text, original JS, MIT license, README quality.
 - **Frustrations:** Sites that only work at desktop width. Creative additions that turn out to be a copied library demo.
 - **What she needs from this site:** A layout that holds up at 390px, a clearly identified creative addition, and an honest statement of what was generated.
-

3. User stories

Written as stories with acceptance criteria, so each one is testable.

Epic A — Establish identity fast

A1. *As Priya, screening quickly, I want the candidate's name, current role and availability without scrolling, so I can triage in seconds.*

- Given a 1440×900 desktop viewport
- When the homepage loads
- Then the name, a one-sentence summary, and "Available ... starting Summer 2027" are all visible above the fold
- And an email link is reachable in one click

A2. *As Priya, I want to know where the candidate studies and when they graduate, so I can match them to a requisition.*

- Given the homepage
- When I reach the Education section
- Then each degree shows institution, degree name, expected date and location

Epic B — Show how I actually think

B1. *As Daniel, I want to see the candidate's architectural thinking, not just a list of technologies.*

- Given the agent board on the homepage
- When I make a move
- Then three agents each report a different recommended reply with a score
- And the coordinator states how many of them it agreed with
- So the disagreement itself is the visible output

B2. *As Daniel, I want to know the chess engine is real and not a library.*

- Given the repository
- Then `js/chess/engine.js` contains the move generator
- And the README publishes perf node counts against known values

B3. *As Meera on a phone, I want to use the board without a mouse.*

- Given a touch device or keyboard-only navigation
- When I Tab to a square and press Enter
- Then that piece is selected and its legal destinations are marked
- And every square announces its name and occupant to a screen reader

B4. *As anyone, I want to undo a bad move rather than restart.*

- Given at least one completed exchange
- When I press "Take back"
- Then my move and the coordinator's reply are both undone

B5. *As Daniel, I want to find where a given skill was used.*

- Given the skill constellation on the Projects page
- When I select "Kubernetes"
- Then every role and project using it is listed, and unrelated nodes dim

Epic C — Depth on demand

C1. *As Daniel, I want longer write-ups than a résumé bullet, so I can prepare questions.*

- Given the Projects page
- Then each entry has a summary plus specific detail points
- And publications are listed with venue and status

Epic D — Work everywhere, for everyone

D1. *As Meera on a 390px phone, I want a usable layout.*

- Given a 390px viewport
- Then the page never scrolls horizontally
- And the navigation collapses behind a labelled Menu button
- And the constellation still renders and responds

D2. *As a keyboard user, I want to skip repeated navigation.*

- Given focus at the top of the document
- When I press Tab once
- Then a "Skip to main content" link appears and is focusable

D3. *As a screen reader user, I want the graph not to be a dead end.*

- Given the canvas
- Then it exposes `role="img"` and a description saying the same nodes exist as buttons below

D4. *As someone with vestibular sensitivity, I don't want a graph animating at me.*

- Given `prefers-reduced-motion: reduce`
- Then the layout is computed without animating and painted once

Epic E — Academic honesty

E1. *As the grader, I want to know exactly which parts were AI-generated.*

- Given the "Built with AI" page
 - Then it names the model and version, lists what was asked, separates generated from hand-authored content, and records where the model was wrong
-

4. Information architecture

```
index.html ..... Home
├── Hero ..... name, one-line pitch, contact, availability
├── Agent chess board .. the creative addition
├── Experience ..... reverse-chronological timeline (4 roles)
├── Education ..... 2 degrees
├── Toolkit ..... grouped skill lists
projects.html ..... Projects + Publications + skill constellation
built-with-ai.html ..... GenAI disclosure (the AI-generated page)
```

All three pages share one persistent header and footer. Content is defined once in `js/data.js`; the constellation and the rendered sections both read from it, so a role is never described in two places.

5. Design mockups

Wireframes produced before implementation. Final screenshots follow in section 8.

5.1 Home — desktop (≥ 992px)

[AP] Ayush PatelHomeProjectsBuilt with AIGitHub|sticky

BOSTON, MA

Ayush Nimeshkumar Patel<- h1, clamp()
MS CS at Northeastern, currently building control-plane
services at AWS. Multi-agent AI systems and the
infrastructure that makes them reliable.

(See my work) (email) (LinkedIn) (GitHub)<- pill links

| Available for full-time roles starting Summer 2027<- accent rule

SKILL CONSTELLATION

A résumé tells you someone knows Kubernetes. It rarely tells
you where they used it.

Databricks

DNS Agents

Python

Pit Wall

Rand-PC

MCP

5G Testbed

Pibit

● Skill

● Experience

● Project

● Publication

PICK A NODE

(AWS)(Computer Vision)(Databricks)
(Deep RL)(DynamoDB)(Java)(Python)
(Kubernetes)(LangChain)(MCP) ...
(Clear selection)

7 cols

Python

7 PLACES I HAVE USED IT

- DNS Anomaly Detection
- 5G Spectrum Testbed
- Pit Wall

5 cols

EXPERIENCE

- SEP 2026 – PRESENT
Software Development Engineer Intern
Amazon Web Services · Boston, MA
Designing S3 endpoint configuration for Transfer Family...
- JUN 2026 – AUG 2026
AI Engineer Intern ...

EDUCATION

TOOLKIT

© 2026 Ayush Patel · Built for CS5610 · Home Projects Source

5.2 Home — mobile (< 768px)

Single column throughout. Nav collapses; the constellation keeps its full interaction but shortens to 340px tall; the detail panel stacks under the chips.

[AP] Ayush Patel (Menu)

BOSTON, MA
 Ayush Nimeshkumar Patel
 MS CS at Northeastern,
 currently building...
 (See my work)
 (email) (LinkedIn)
 | Available Summer 2027

SKILL CONSTELLATION

●Skill ●Exp ●Proj

PICK A NODE

(AWS)(CV)(Databricks)

(Deep RL)(DynamoDB) ...

Python
 7 PLACES

EXPERIENCE

● SEP 2026 – PRESENT
 | SDE Intern
 | Amazon Web Services

Nav open state:

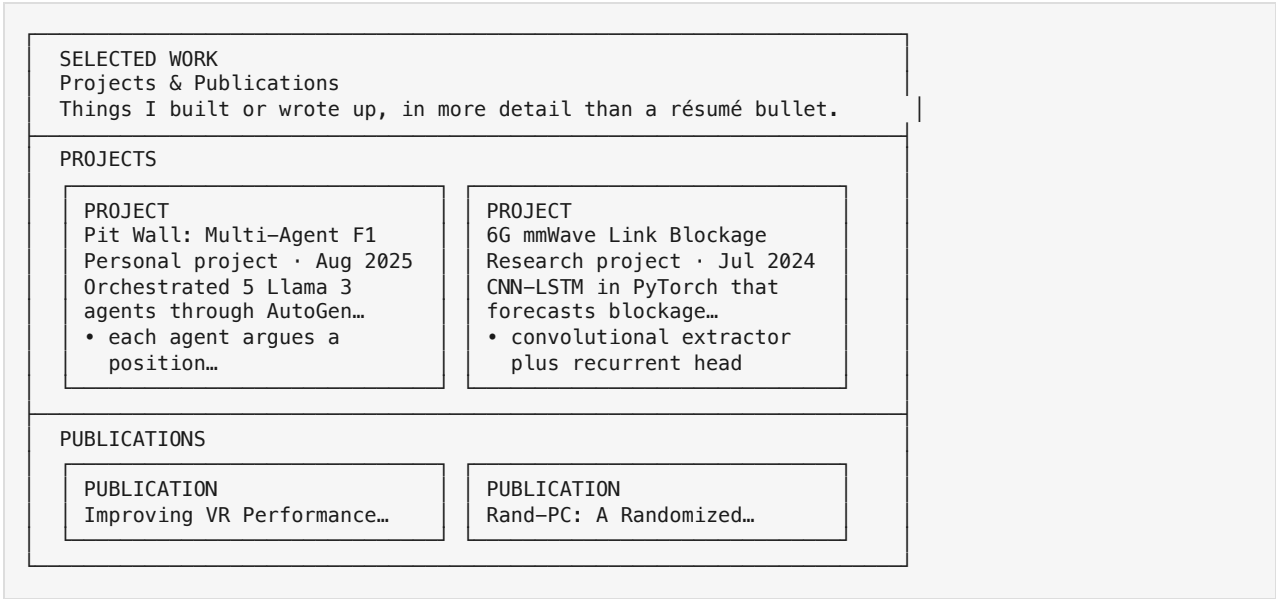
[AP] Ayush Patel (Menu)

Home
 Projects
 Built with AI
 GitHub

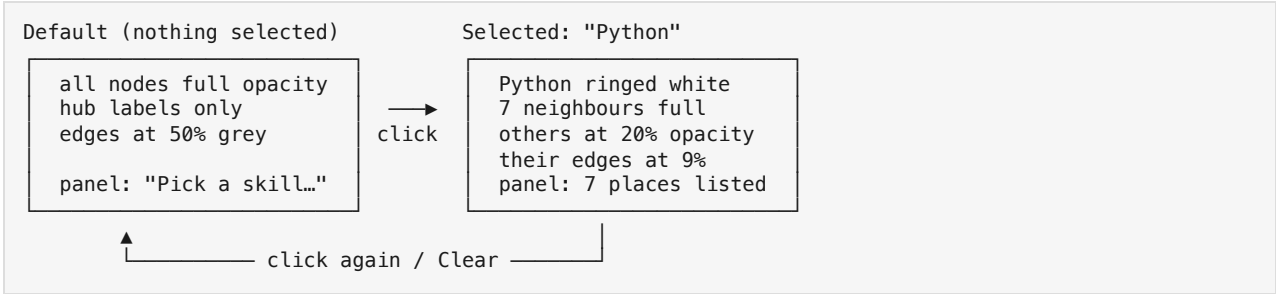
aria-expanded="true"

5.3 Projects — desktop

Two-column card grid (col-12 col-lg-6), collapsing to one column on mobile.



5.4 Constellation interaction states



6. Design system

Palette

Borrowed from chess.com, because the board is the centrepiece and the rest of the page should look like it belongs to the same product. Warm near-black rather than blue-black, so the board's greens sit naturally on it, with a single bright green carrying every call to action.

Token	Value	Use
--ink	#262421	Page background (warm black)
--surface	#312e2b	Cards, nav, status panel
--surface-raised	#3d3a37	Pills, chips, buttons

Token	Value	Use
--line	#4a4642	All borders and rules
--text	#f1efed	Body text
--text-muted	#a9a29b	Secondary text
--accent	#81b64c	Links, primary buttons
--board-light	#eeeeed2	Light squares
--board-dark	#769656	Dark squares
--board-*--active	#f6f669 / #baca2b	Last-move highlight
--warn	#e0a63c	"No preference" notices

Only one hue does the accent work. Everything that is interactive is green; nothing decorative is. That is what keeps a dark page from turning muddy.

Type and spacing

System font stack, so there is no web-font round trip. The h1 uses `clamp(2.1rem, 6vw, 3.4rem)` to scale without breakpoints. Body copy is capped at 58–70ch for readability. Spacing is a six-step scale (`0.25/0.5/1/1.5/2.5/4rem`) exposed as custom properties, so vertical rhythm stays consistent.

CSS conventions

- Every element is targeted by **class**, never by id, and never by bare tag beyond base resets.
- **Zero !important** in the stylesheet (verified: `grep -c '!important' → 0`).
- Layout uses Bootstrap's grid plus CSS flexbox and grid. One deliberate override: `.page-wrap` `.row` zeroes Bootstrap's negative gutters, because `.page-wrap` is not a Bootstrap `.container` and the negative margins otherwise cause horizontal overflow on mobile.

Accessibility decisions

Concern	Decision
Keyboard access to the graph	Every node is also a real <code><button></code> with <code>aria-pressed</code>
Screen readers	Canvas is <code>role="img"</code> with a label pointing at the button list
Skip navigation	<code>.skip-link</code> , off-screen until focused

Concern	Decision
Focus visibility	One shared :focus-visible outline, 3px, offset 3px
Motion sensitivity	prefers-reduced-motion settles the layout without animating
Semantic controls	Buttons are <button>, links are <a> — no clickable <div>
Images	Every carries a real alt

7. The creative addition: an agent committee that plays chess

Why this and not a honeycomb grid

The assignment's own example is a honeycomb image grid, so that is the one thing guaranteed to appear in other submissions. More importantly, a decorative grid says nothing about how I actually work.

At AT&T I built a multi-agent system where one orchestrator coordinates three specialist sub-agents over MCP, and the useful signal there was never the final answer — it was **where the specialists disagreed**. This widget is that architecture, shrunk to something you can play with in ten seconds.

Chess is an honest demonstration domain because the objectives genuinely conflict: grabbing a free pawn can wreck your king. A committee that always agrees would teach nothing.

The engine

Written from scratch in `js/chess/engine.js`, with no chess library:

- **0x88 board representation** — off-board detection is a single bit test
- **Fully legal move generation** — castling (including through and out of check), en passant, promotion, pins, checkmate and stalemate
- **Make / unmake** with a history stack, so the search never copies the board
- **Alpha-beta negamax** with capture-first move ordering

The agents

Agent	Objective	Deliberately ignores
Material	Piece values only	Position entirely
Space	Central control, piece activity	Material entirely
King safety	Shelter, exposure, distance from home rank	Material entirely

Each searches independently to depth 3 with its own evaluation. A coordinator then searches with a weighted blend (1.0 / 0.35 / 0.55) and picks the move. The panel reports how many agents the coordinator agreed with.

Honest reporting of indifference

A pure material agent has **no opinion** in a quiet opening — every move scores zero. Rather than present an arbitrary pick as a recommendation, the agent says so.

This was a bug first. The original test was "top move ties with second", which wrongly flagged an agent as indifferent when its two *best* moves tied — the agent that had just found a way to win a pawn was reported as having no preference. Real indifference is when every move scores the same.

How correctness is established

A move generator can be subtly wrong in ways no amount of playing by hand will reveal. So the engine is verified with **perft** — counting leaf nodes of the move tree to a fixed depth against published values:

Position	Depth	Expected	Result
Start position	4	197,281	pass
Kiwipete	3	97,862	pass
En passant / pins	4	43,238	pass
Promotions	3	9,467	pass
Position 5	3	62,379	pass

The engine was perft-verified **before any interface was built on top of it**.

Accessibility

Every square is a real `<button>` in a grid with an `aria-label` naming the square and its occupant, so the board is fully keyboard-operable and legible to a screen reader. Pieces are Unicode glyphs; there is no sprite sheet.

Performance is roughly 12,000 positions in ~50ms, so no web worker is needed.

7b. Second interactive piece: Skill Constellation

Moved to the Projects page. A force-directed graph wiring all 23 skills and work items together: pick a skill and every place I used it stays lit while the rest dims.

Layout uses Fruchterman–Reingold: repulsion k^2/d , attraction d^2/k , ideal distance $k = 0.62 \cdot \sqrt{(\text{area} / \text{nodeCount})}$ derived from canvas size and node count, and a cooling temperature so it converges. Labels are drawn in a second pass with rectangle-collision detection.

8. Final implementation

Home — hero

**Ayush Patel**

HomeProjectsBuilt with AIGitHub

BOSTON, MA

Ayush Nimeshkumar Patel

MS Computer Science student at Northeastern, currently building control-plane services at AWS. I spend most of my time on multi-agent AI systems, the infrastructure that makes them reliable, and the networks they run over.

[See my work](#)ayushnpatelworks@gmail.com[LinkedIn](#)[GitHub](#)

Available for full-time software engineering roles starting **Summer 2027**.



At the AT&T Innovation Center R&D Labs, Middletown NJ, during my Network CTO internship.

PLAY THE AGENT COMMITTEE

At AT&T I built a system where one orchestrator coordinates three specialist agents, and the useful signal was never the final answer — it was *where the specialists disagreed*. Chess is a clean way to show that, because the objectives genuinely conflict: taking a free pawn can wreck your king.

You are White. Make a move and three agents each search the position with a different objective, then a coordinator reconciles their votes. The engine underneath

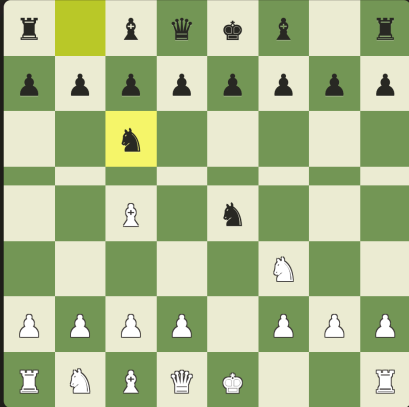
The agent board mid-game

**Ayush Patel**

[Home](#) [Projects](#) [Built with AI](#) [GitHub](#)

conflict: taking a free pawn can wreck your king.

You are White. Make a move and three agents each search the position with a different objective, then a coordinator reconciles their votes. The engine underneath — move generation, castling, en passant, promotion, alpha-beta search — is written from scratch and verified with `perft`.



New game

Take back

The coordinator played **Nc6** — 1 of 3 agents agreed.
11,201 positions searched in 42ms

Material

ADOPTED

Counts what is on the board and nothing else.

Nc6 +200

Space

Rewards central control and active pieces, ignores material.

b5 +35

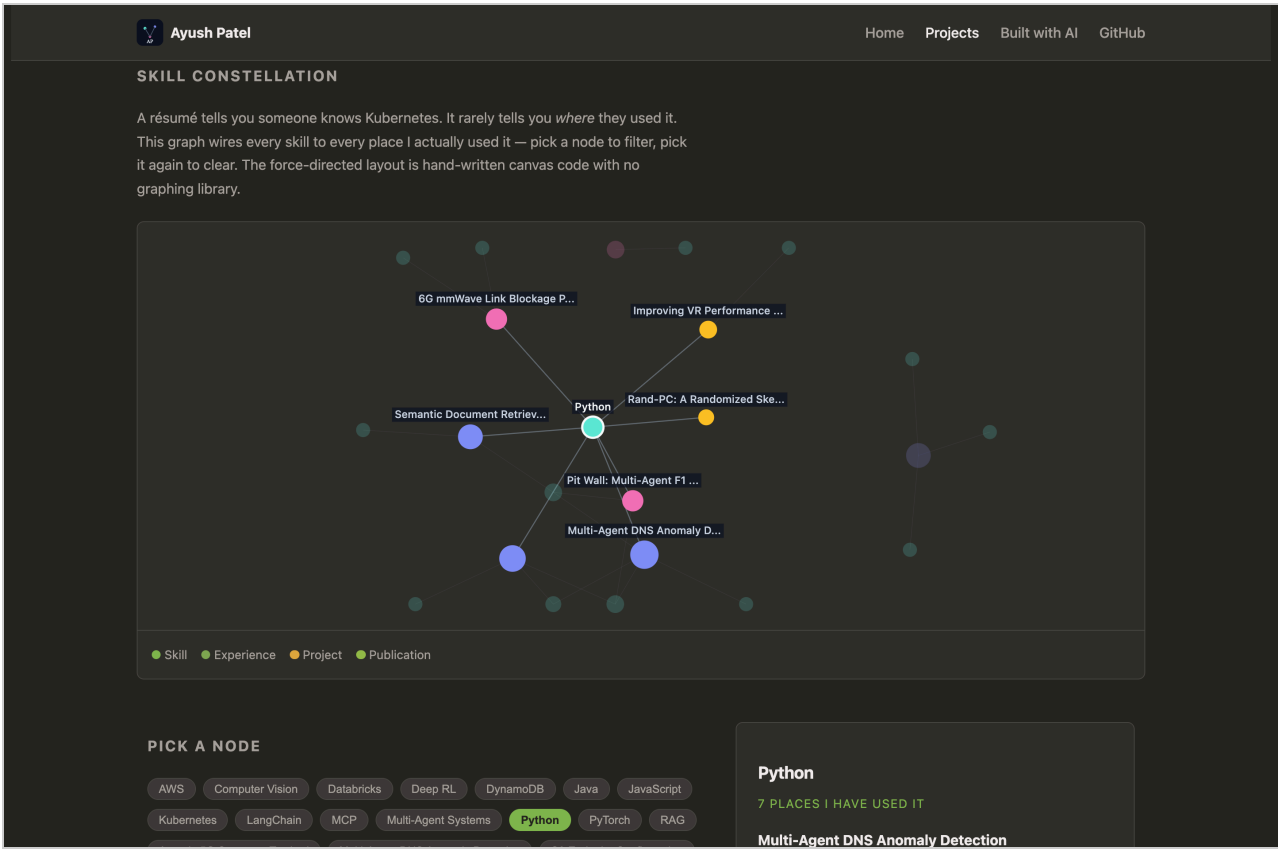
King safety

Cares only about shelter and exposure around both kings.

Rg8 +34

Each agent reports its own recommendation and score; the one the coordinator adopted is outlined in green.

Skill constellation, Projects page



Mobile



Ayush Patel

Menu

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9. Verification

Check	Method	Result
W3C validity	validator.w3.org/nu on all 3 pages	0 errors
ESLint (class config)	npx eslint .	0 errors
Prettier	npx prettier --check	clean
Console errors	Headless Chrome, all 3 pages	none
Images have alt	DOM audit	0 missing
Horizontal scroll at 390px	scrollWidth vs innerWidth	none
Keyboard filtering	Tab + Enter on node list	works
!important count	grep -c on the stylesheet	0

